Phase 3 - Team 57

Relational Model (Stage A)

1. Stage A explanation

The assigned EER diagram became **29 relations** with **36 foreign keys**. Primary keys are underlined in the diagram. Primary keys are written below as `PK(...)`.

1.1 Entity types

Strong entity types each became one relation with all simple attributes. Composite attributes were flattened into their components. (Since we're flattening composite attributes that's for stage B as well)

Relation	PK	Notes
RACE_EVENT	RaceEventID	
SCORING_SCHEME	ScoringSchemeID	`VersionLabel` is the alternate key (UNIQUE).
COMPETITION_EDITION	CompetitionEditionID	Composite `Period` became `StartDate`, `EndDate`.
ORGANISATION	OrganisationID	
PERSON	PersonID	
VENUE	VenueID	Composite `Location` became `CountryCode`, `City`, `Region`.
COURSE	CourseID	`VenueID` NOT NULL, because R2 is (1,1) on the course side.
COURSE_RECORD	CourseRecordID	Composite `RecordCategory` became `Division`, `AgeGroupLabel`.
RANKING_PUBLICATION	RankingPublicationID	Composite `RankingWindow` became `WindowStartDate`, `WindowEndDate`. They were renamed because `StartDate` and `EndDate` are ambiguous.

Weak entity types each include the owner's PK as a foreign key. The PK is the owner's key plus the partial key.

- **RACE_EDITION** is owned by RACE_EVENT through R1. `PK(RaceEventID, OccurrenceNo)`. Composite `Schedule` became `StartDate`, `EndDate`, and `RankingDate` is a plain column.
- **RESULT_SUBMISSION** is owned by RACE_EDITION through R16. `PK(RaceEventID, OccurrenceNo, SubmissionNo)`. This is the three-part key from the diagram.
- **ANNUAL_MEMBERSHIP** is owned by REGISTERED_ATHLETE through R9. `PK(PersonID, MembershipYear)`.

Derived attributes not stored: `Age` (PERSON), `DurationDays` (RACE_EDITION) and `MembershipValid` (ANNUAL_MEMBERSHIP). They can be computed from `BirthDate`, `StartDate` / `EndDate` and `ValidFrom` / `ValidUntil` / `ConfirmationStatus`.

1.2 Specialization (textbook Option 8A, one relation per class)

The EER has PERSON ⊃ ATHLETE ⊃ REGISTERED_ATHLETE. Both specializations are partial and have one subclass each. We used **Option 8A (a relation for the superclass and one for each subclass)**:

- ATHLETE has `PK(PersonID)`, and `PersonID` is an FK to PERSON.
- REGISTERED_ATHLETE has `PK(PersonID)`, and `PersonID` is an FK to ATHLETE. `TWAAthleteID` is UNIQUE NOT NULL as the alternate key.

Reason: both are partial, so a single table would be full of NULLs. Each subclass also has its own relationships (R4, R13 use ATHLETE; R9, R11 use REGISTERED_ATHLETE).

1.3 Relationships

EER relationship	Mapped as
R1 HAS_EDITION, R16 HAS_SUBMISSION, R9 HAS_MEMBERSHIP (identifying)	Owner's PK in the weak relation, as above.

EER relationship	Mapped as
R2 HAS_COURSE (VENUE 1:N COURSE)	FK `COURSE.VenueID`, NOT NULL.
R12 HAS_RECORD_CLAIM (COURSE 1:N COURSE_RECORD)	FK `COURSE_RECORD.CourseID`, NOT NULL.
R13 HELD_BY (ATHLETE 1:N COURSE_RECORD, optional)	FK `COURSE_RECORD.HolderPersonID` → ATHLETE, nullable.
R10 SCORED_USING	FK `RACE_EDITION.ScoringSchemeID`, nullable.
R15 HELD_AT	FK `RACE_EDITION.VenueID`, nullable.
R17 PROVIDED_BY	FK `RESULT_SUBMISSION.ProvidedByOrganisationID` → ORGANISATION, nullable.
R14 PREVIOUS_EDITION (recursive, 1:1)	FK `(PrevRaceEventID, PrevOccurrenceNo)` → RACE_EDITION, nullable and UNIQUE.
R3 OFFERS (M:N)	Relation OFFERS, `PK(RaceEventID, OccurrenceNo, CourseID)`, with `EvaluationSelection` and `SelectionEvidence`.
R5 INCLUDED_IN (M:N)	Relation INCLUDED_IN, `PK(RaceEventID, OccurrenceNo, CompetitionEditionID)`, with `RoleLabel` and `InclusionNote`.
R6 ORGANISES (M:N)	Relation ORGANISES, `PK(OrganisationID, RaceEventID, OccurrenceNo)`.
R7 AFFILIATED_WITH (M:N)	Relation AFFILIATED_WITH, `PK(PersonID, OrganisationID)`.
R8 ASSISTS_EDITION (M:N)	Relation ASSISTS_EDITION, `PK(PersonID, RaceEventID, OccurrenceNo)`.
R11 LISTED_IN (M:N)	Relation LISTED_IN, `PK(PersonID, RankingPublicationID)`, with `PublishedDivision`, `CountryAtPublication`, `PublishedRank`, `PublishedPoints`. `PersonID` is an FK to REGISTERED_ATHLETE.
R4 ACHIEVES_RESULT (ternary) RACE_EDITION–COURSE–ATHLETE	Relation ACHIEVES_RESULT with FKs to all three: `(RaceEventID, OccurrenceNo)` → RACE_EDITION, `CourseID` → COURSE, `PersonID` → ATHLETE. `PK(RaceEventID, OccurrenceNo, CourseID, PersonID)`. Carries `CompetitionDivision`, `CountryAtRace`, `FinishStatus`, `VerificationStatus`, `DisputeStatus`, `SourceURL`,

EER relationship	Mapped as
	`OverallPlace`, `FinishingTimeSeconds`, `Points`, plus the `SourceCitation` columns (see 1.4).

1.4 Multi-valued attributes (all Option (b), separate relation)

No requirement in the assigned document gives a fixed maximum number of values for any of these attributes, so Option (a) was not used for any of them.

Multi-valued attribute	New relation	PK	One-line reason
`MatrixEntries` (SCORING_SCHEME)	SCORING_MATRIX_ENTRY	(ScoringSchemeID, Factor, PlaceFrom)	Unbounded composite with 4 sub-attributes; the non-overlap rule per factor makes `PlaceFrom` unique within (scheme, factor).
`AlternativeNames` (VENUE)	VENUE_ALTERNATIVE_NAME	(VenueID, AlternativeName)	Number of names not fixed.
`ContactEmails` (PERSON)	PERSON_CONTACT_EMAIL	(PersonID, ContactEmail)	Number of emails not fixed.
`SourceRows` (RESULT_SUBMISSION)	SOURCE_ROW	(RaceEventID, OccurrenceNo, SubmissionNo, SourceRowNo)	Unbounded composite with 4 sub-attributes; FK to RESULT_SUBMISSION.
`NationsStandings` (RANKING_PUBLICATION)	NATIONS_STANDING	(RankingPublicationID, CountryCode)	One row per country per publication; composite.
`AdditionalPlacings` (R4)	ADDITIONAL_PLACING	(RaceEventID, OccurrenceNo,	Unbounded composite on a relationship; FK to ACHIEVES_RESULT.

Multi-valued attribute	New relation	PK	One-line reason
		CourseID, PersonID, ClassificationLabel)	
`CurrentRoleTitles` (R7)	AFFILIATION_ROLE_TITLE	(PersonID, OrganisationID, RoleTitle)	Number of titles not fixed; FK to AFFILIATED_WITH.
`Responsibilities` (R8)	ASSISTS_EDITION_RESPONSIBILITY	(PersonID, RaceEventID, OccurrenceNo, Responsibility)	Number of responsibilities not fixed; FK to ASSISTS_EDITION.

1.5 Not drawn as relations

- No union types (categories) exist in the assigned diagram, and no 1:1 relationships other than R14.
- Constraints that the EER could not express (the R4 pair must exist in OFFERS, acyclic PREVIOUS_EDITION chains, venue consistency between R3 and R15, immutability of published rankings) are not represented in the relational diagram. They are left for Phase 4 as CHECK constraints or triggers.

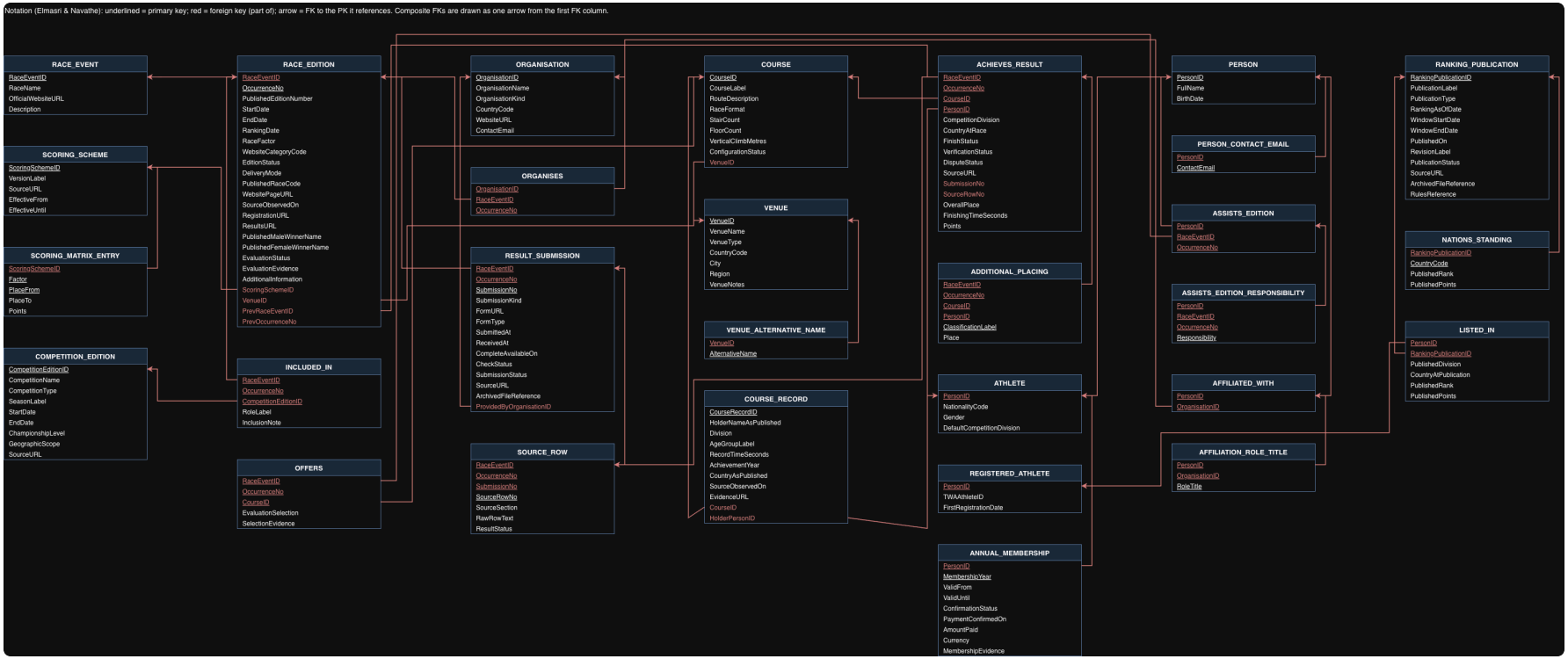
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2. Changes made to the assigned EER diagram

1. **Added an FK from ACHIEVES_RESULT to SOURCE_ROW.** The EER has `SourceCitation` as a composite attribute of R4 whose components are `(RaceEventID, OccurrenceNo, SubmissionNo, SourceRowNo)`. `SourceRows` is a multi-valued attribute of RESULT_SUBMISSION, so the EER cannot draw a line between them. Since SOURCE_ROW became a relation, we mapped the citation as the composite FK `(RaceEventID, OccurrenceNo, SubmissionNo, SourceRowNo)`. `RaceEventID` and `OccurrenceNo` are reused from the R4 key, so only `SubmissionNo` and `SourceRowNo` are extra nullable columns. No information was lost; the FK only makes the link enforceable.

No other changes were made. The assigned team also listed no changes beyond its evaluation.

3. Stage A Snapshot



https://tinyurl.com/phase3-relationalschema

Notation: Elmasri & Navathe relational schema. Underlined = primary key. Red attribute names = part of a foreign key. An arrow goes from the FK to the PK it references. A composite FK is drawn as one arrow, starting at its first FK column.

Stage B

NO CONVERSION NEEDED.

Since all attributes are atomic and no composite or lists are in any columns it's already 1NF.
Checked every relation in Section 1.4 (Multi-valued attributes) to see if there could be list in a column.

Stage C

NO CONVERSION NEEDED

Checked relations ACHIEVES_RESULT, ANNUAL_MEMBERSHIP, LISTED_IN, SCORING_MATRIX_ENTRY, and OFFERS

ANNUAL_MEMBERSHIP requires both PersonID and MembershipYear because there will be multiple tuple sets one for each year
Similarly checked the rest of the relations

Also checked the weak entity relations (refer section 1.1 latter half)

Also the multi-value attribute tables (refer section 1.4)

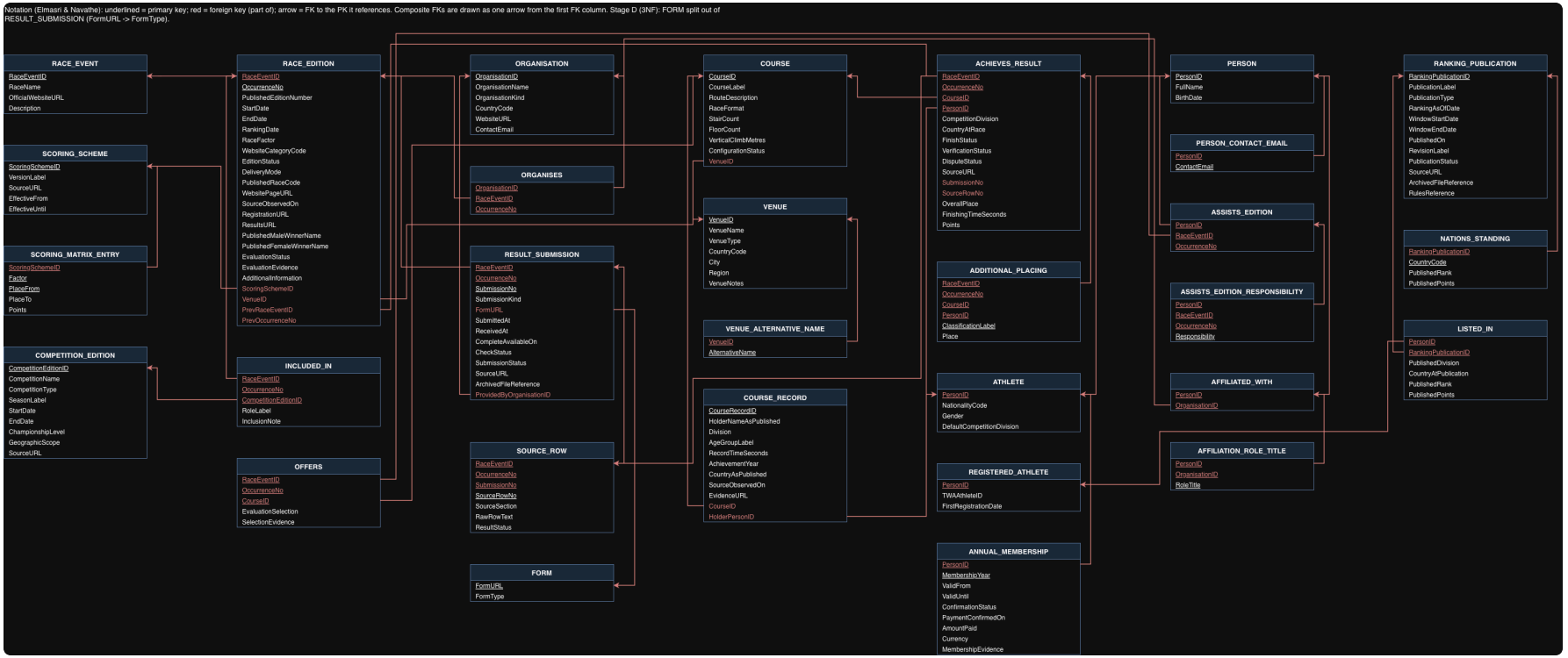
Stage D

RESULT_SUBMISSION had the transitive dependency SubmissionNo, RaceEventID, OccurrenceNo → FormURL → FormType.

FormURL is not a key and FormType is not prime, so FORM(FormURL, FormType) was created and RESULT_SUBMISSION.FormURL became a foreign key to it

(Assumption: Each Results Form URL identifies exactly one form type (standard, Factor-40 or other), based on the Ranking page and not multiple)

Functional Dependency - FormURL $\Longrightarrow$ FormType



https://tinyurl.com/phase3-threeNF

GenAI usage statement

We used Claude to help read the assigned team's Phase 2 PDF, to draft the mapping from the EER diagram to relations, and to finalise the wording of this report.